



Self-powered heating strategy for lithium-ion battery pack applied in extremely cold climates

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ABSTRACT

Serious performance loss of lithium-ion batteries at subzero temperatures is the major obstacle to promoting battery system in cold regions. This paper proposes a novel heating strategy to heat battery from extremely cold temperatures based on a battery-powered external heating structure. The strategy contains two stages: preheating process for battery cold-start, and temperature holding process for battery temperature control after preheating. The strategy switches from the preheating to the temperature holding according to the power capability of battery pack. Both the electrical and thermal behaviors of battery pack are described based on equivalent circuit models, and the state-of-charge and state-of-power of battery pack are co-estimated online to support the implementation of the strategy. Systematic battery-in-the-loop tests are conducted for a series-connected battery pack to verify the effectiveness of the strategy at $-40\text{ }^{\circ}\text{C}$. Experimental results show that the battery pack can not only be heated from $-41.4\text{ }^{\circ}\text{C}$ to $-14.0\text{ }^{\circ}\text{C}$ in 371 s by consuming 6.1% of nominal capacity without being over-discharged, but also output 64.81% and 66.59% of nominal energy for loading the modified dynamic stress test cycles and the modified federal urban driving schedule cycles respectively after preheating.

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1. Introduction

In the past decade, battery energy storage systems (BESSs) have been widely utilized in various promising fields, such as electric vehicles (EVs) [1], fuel cell vehicles [2] and off-grid power station [3]. Lithium-ion batteries (LIBs) play the key role in BESS because of their high energy density and long lifetime [4]. However, the LIBs suffer from serious performance loss at low temperatures [5,6], which is a natural barrier that prevents the BESSs from being promoted in cold regions. Moreover, the unconstrained low-temperature operations may cause irreversible damages on LIBs, such as occurring lithium-ion deposition [7]. Experimental investigations reveal that both the power capability and the available capacity of LIBs can be restored at low temperatures after being properly heated up [8], and the potential damages can be avoided regardless of the operating temperature when batteries discharge in the specified voltage range [9]. Therefore, the appropriate battery heating is the key to the efficient application of commercial LIBs at low temperatures. At present, the battery cooling attracts

more researchers compared with the heating, because working long hours at excessively high temperatures will pose a serious threat to battery life [10]. For most of BESSs in market, the heating strategy is usually integrated with the cooling strategy into battery management system (BMS) [11]. Considering the limited heating performance [12] and the demand of extra power supply [13], the existing cooling/heating integrated strategies are ineffective at extremely cold temperatures (ECTs) below $-40\text{ }^{\circ}\text{C}$. Hence, designing the specialized heating strategy is crucial to solve the problem of battery performance degradation at ECTs.

1.1. Reviews of heating strategy for lithium-ion batteries

According to the features of heating current, the existing strategies can be divided into alternating current (AC) heating [14–20] and direct current (DC) heating [21–23, 25–28, 30–32]. Depending on whether the dominant heating source is outside or inside battery, the strategies can also be separated into internal heating [14–23] and external heating [25–29]. In addition, some strategies can be called as battery-powered heating or self-heating [18–20, 22–25], if the heating power is supplied by batteries themselves.

Recently, the AC heating attracts great attention due to its

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Nomenclature		T_{avg}^{pack}	Average temperature of pack (°C)
Symbol		T_j^{hf}	Temperature of j -th heating film (°C)
		C_a	Nominal capacity of battery cell (Ah)
V_{cell}	Terminal voltage of i -th cell (V)	Abbreviation	
$V_{oc,i}^{cell}$	Open-circuit voltage of i -th cell (V)	AC	Alternating current
$V_{p,i}^{cell}$	Polarization voltage of i -th cell (V)	ARLS	Adaptive recursive least squares
V_t^{pack}	Terminal voltage of pack (V)	BESS	Battery energy storage system
S_{cell}^{cell}	SoC of i -th battery cell (%)	BMS	Battery management system
S_{oc}^{pack}	SoC of battery pack (%)	BTS	Battery test system
$R_{o,i}^{cell}$	Ohmic resistance of i -th cell (Ω)	DAQ	Data acquisition
$R_{p,i}^{cell}$	Polarization resistance of i -th cell (Ω)	DC	Direct current
$C_{p,i}^{cell}$	Polarization capacitance of i -th cell (F)	DCR	Direct-current resistance
$R_{h,i}^{pack}$	Thermal resistance of pack (K/W)	DST	Dynamic stress test
C_{cell}^{cell}	Thermal capacity of pack (J/K)	EECM	Electrical equivalent circuit model
T_{tar}^{pack}	Setting temperature of heating films (°C)	EKF	Extended Kalman filter
T_{box}	Temperature in battery box (°C)	ECT	Extremely cold temperature
T_{amb}	Temperature in test chamber (°C)	EV	Electric vehicle
Δt	Sample interval (s)	FUDS	Federal urban driving schedule
I_{pack}^{tot}	Total output current of pack (A)	HF	Heating film
I_{pack}^{heat}	Output heating current of pack (A)	LIB	Lithium-ion battery
$P_{pack}^{heat,max}$	Permissible heating power of pack (W)	MAE	Mean absolute error
$P_{load,max}^{pack,ls}$	Maximum power of loads (W)	OCV	Open-circuit voltage
$P_{pack,ls}^{max}$	Short-term SoP of pack (W)	PWM	Pulse width modulation
$P_{pack,ll}^{max}$	Long-term SoP of pack (W)	RMSE	Root mean square error
Q_{os}^{pack}	External heat generation rate (W)	RLS	Recursive least squares
Q_{pack}^{pack}	Internal heat generation rate (W)	SoC	State-of-charge
$P_{dc/dc}^{in}$	Input power of DC/DC converter (W)	SoP	State-of-power
$P_{out}^{dc/dc}$	Output power of DC/DC converter (W)	SSR	Solid-state relay
$\eta_{dc/dc}$	Efficiency of DC/DC converter (%)	TCM	Temperature control module
T_i^{cell}	Average temperature of i -th cell (°C)	TECM	Thermal equivalent circuit model

advantage of heating battery from inside efficiently. Early in 2004 Stuart et al. [14] indicated that loading AC can preheat rechargeable batteries based on the Joule effect occurring on battery internal resistance, and proved that both the low frequency current at 60 Hz and the high frequency current above 10 kHz can be utilized for heating. Typically, the low-frequency AC heating methods require an extra power source to control the charge and discharge of battery, and the strategies for this kind of method focus on optimizing the amplitude and frequency of current to improve heating performance. Zhu et al. [15] investigated the effect of low-frequency AC preheating on the state-of-health of large-capacity prismatic LIB, and showed that using the AC pulse current at 2 C-rate and 0.5 Hz can warm the cell up from -25°C to 0°C in 30 min. Guo et al. [16] proposed an echelon preheating strategy which can raise the temperature of 18,650 cell from -20°C to 10°C in 13.7 min with the AC at 10 Hz. Li et al. [17] proposed an internal preheating strategy to load sinusoidal AC for heating, and they revealed that using the sinusoidal AC at 1.9 C-rate and 1585 Hz can heat the 18,650 cell up from -20°C to 0°C within 9 min.

Given the fact that providing the extra power source is inconvenient in some situations, it is another feasible way to achieve AC heating by designing the battery-powered heater based on high-frequency electrical circuit. Shang et al. [18] proposed a preheating strategy based on a resonant switched capacitor heater, and created a sinusoidal current that preheated the 18,650 cell from -20°C to 0°C in 7.4 min. In Ref. [19], another high-frequency AC heater was developed for 18,650 cells based on buck-boost circuit. Jiang et al. [20] utilized a resonance circuit to generate the current composed of AC and DC for heating, and preheated a prismatic battery pack from -20.8°C to 2.1°C within 10 min. Above-

mentioned high-frequency AC heaters can be integrated into BMS, but they face the problem of high energy loss in heating circuit.

Similar to the AC heating, the internal DC heating can be implemented by controlling the output current of battery according to the designed strategy. Ruan et al. [21] presented a model-based strategy to optimize the output voltage of battery when loading continuous DC for internal self-heating, and verified that the capacity loss of the battery without being over-discharged was less than 2% after preheating 500 times. Qu et al. [22] developed a self-heating circuit and presented a strategy to make battery discharge pulse DC, and they utilized the generated pulse current to warm an 18,650 cell from -10°C to 10°C within 3 min. To fully exploit the battery energy consumed by the transistor in self-heating circuit, Xu et al. [23] designed a structure with two heat sources for prismatic battery and presented a dual mode strategy for heating control.

However, it is difficult to accurately control the battery temperature when using the AC heating or the internal DC heating. Ref. [24] revealed that the thermal management with well-designed external structure was conducive to the uniform temperature distribution in large-scale prismatic battery pack. Therefore, the controllable external heating elements, such as the electrically heated film, are widely used in engineering. In Ref. [25], a battery-powered strategy was presented based on an external heating structure equipped with heating film (HF), which can preheat a prismatic battery pack from -40°C to 0°C within 10 min. Min et al. [26] developed a charging-heating combined strategy, and they warmed up the 18,650 cell externally during charging process by controlling both the power and temperature of HF. Wang

et al. [27] presented a structure composed of HF and heat pipe array, and utilized the structure to preheat a prismatic battery cell from -30°C to 0°C within 20 min. To decrease the temperature differences among in-pack cells during preheating, He et al. [28] presented a structure of using HF with phase change materials, but they encountered the problem of slow heating speed. Ling et al. [29] proposed a fast-heating strategy to actively release the energy in supercooled phase change materials. The major problem is that it is difficult to store energy for phase change materials at ECTs.

Moreover, in Ref. [30], the all-climate battery structure was first developed by inserting metal foil into the prismatic battery for internal heat generation in cold temperatures. Based on the prototype of all-climate battery, Zhang et al. [31] improved the self-heating structure and proposed a preheating strategy that can increase the battery temperature from -40°C to 0°C within 112 s. Lei et al. [32] established a thermal finite element model of prismatic battery, and compared the temperature uniformities during two preheating processes that utilized the internal self-heating structure in Ref. [31] and the external heating film respectively. Besides, Wu et al. [33] presented a passive strategy for keeping battery warm at subzero temperatures, and utilized the nano-porous aerogel to surround the 18,650 cell for heat insulation. Song et al. [34] proved the necessity of preheating process for the BESSs operating in ambient temperatures ranging from -20°C to 0°C .

Admittedly, the existing strategies have good performance in preheating various LIBs, but the preheating alone is not sufficient to ensure the efficient utilization of battery energy at the ECT of -40°C . After preheating, the temperature difference between battery pack and surrounding air is greater than 40°C . At this time, the heat generated internally from battery pack when loading is far less than heat dissipation, which leads to the rapid decrease of battery temperature. Consequently, the performance of the preheated battery pack can be only temporarily boosted, if the strategy is not formulated to continuously control battery temperature after preheating.

1.2. Motivation and contributions of this paper

To maximize the available capacity and available energy of the battery pack working at ECTs, it is crucial to keep controlling the temperature of the preheated battery pack. Currently, there is still a lack of heating strategy that can fully cover both the preheating and the temperature control after preheating. To accurately control the battery temperature and to reduce the difference among the temperatures of in-pack cells, the external DC heating approach that employs the HF as heating element is utilized in this paper. For achieving the heating control without using extra power source at ECTs, there remains three dominant challenging problems as follows:

- (1) How to control the heating power for rapid preheating battery pack without incurring over-discharge.
- (2) How to switch from preheating to battery normal operation and temperature control.
- (3) How to control the heating temperature for energy-efficient operation after preheating.

In response, this paper proposes a self-powered heating strategy based on an external heating structure. Different from the internal heating, the designed external heating structure facilitates more precise temperature control for battery pack. The strategy contains two processes, namely the preheating and the temperature holding, which are employed for battery cold-start and normal operation respectively. Through inputting the online predicted battery states into the strategy, the battery pack can preheat itself by

utilizing its permissible power without being over-discharged. When the preheated battery pack satisfies the power demand of external loads, the strategy switches from the preheating to the temperature holding, and starts to adaptively control battery pack temperature for improving energy-efficiency during normal operation at ambient temperature of -40°C .

1.3. Paper organization

The remainder of this paper is organized as follows. The self-powered external heating structure and the battery pack modeling are introduced in Section 2. The details of implementing the strategy, as well as the core algorithms of model-based prediction, are described in Section 3. Experimental verification and evaluation of the strategy performance are shown in Section 4. The conclusions are reported in Section 5.

2. Problem formulation and modeling

The heating structure and the control circuit are presented in this section. A series-connected battery pack composed of 4 prismatic cells is selected as the heating object to conduct a case study. To prevent in-pack cells from being over-discharged at -40°C , both the electrical behavior and the thermal behavior of battery pack need to be modeled. Considering the tradeoff between the complexity and the accuracy of real-time computation, the equivalent circuit model is employed in battery modeling.

2.1. Topology configuration of self-powered external heating

An external heating structure is presented in Fig. 1. For most of commercial prismatic cells, thin heating film (HF) is an ideal external heating source. Heat generation depends on Joule effect of the iron-chromium-aluminum alloy wires inside HF. The HFs adhere to both main surfaces of each cell, which permits heat to penetrate cell evenly from both sides and consequently facilitates efficient heat conduction. The details of the tested cell and the HF are listed in Table 1. To reduce heat dissipation, the battery pack is constructed from closely arranged cells.

The battery self-powered heating circuit in Fig. 2 is developed to support heating control. The heating power of battery pack can be controlled precisely by restricting the output power of the DC/DC converter. To avoid the potential problems including local overheating and temperature fluctuation during heating, this paper utilizes the temperatures of HFs (denoted by T_j^{hf} , $j = 1, 2, \dots, 5$) as the feedback in close-loop of temperature control. Then, the PID algorithm can be employed to achieve the temperature control through sending the generated pulse width modulation (PWM) signals to solid-state relay (SSR) module. Since the thermal capacity of battery cell is larger than that of HF, the temperatures of in-pack cells (denoted by T_i^{cell} , $i = 1, 2, 3, 4$) will get close to the controlled temperatures of HFs after a brief delay.

2.2. Electrical dynamic modeling of battery pack and DC/DC converter

Considering that Thevenin model is sufficiently simple and relatively accurate among widely used equivalent circuit models [35], this paper employs Thevenin model to describe the electrical dynamic of battery pack. Fig. 3(a) illustrates the established in-pack cell model which can be expressed in discrete-time form by:

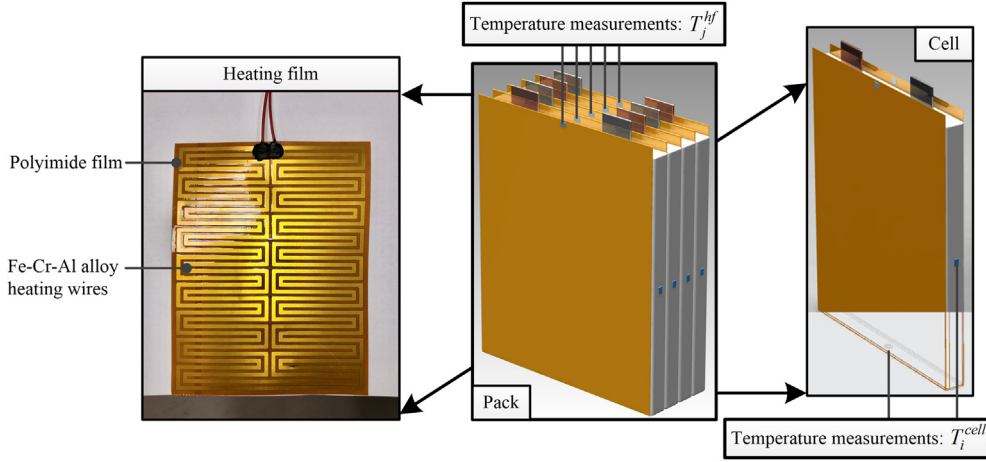


Fig. 1. External heating structure of prismatic battery pack.

Table 1
Details of the tested battery cell and heating film.

Items	Specifications	Values
Battery cell	Electrode materials	Li(NiCoMn)O ₂ /Graphite
	Size (mm)	110 × 95 × 8
	Nominal voltage (V)	3.7
	Nominal capacity (Ah)	10
	Voltage range (V)	3–4.2
Heating film	Film materials	Polyimide/Fe–Cr–Al alloy
	Size (mm)	120 × 90 × 0.2
	Nominal power (W)	24
	Nominal voltage (V)	12

$$S_{oc,i}^{cell}(k) = S_{oc,i}^{cell}(k-1) + \frac{\eta_c \Delta t I_{tot}^{pack}(k)}{3600 C_a} \quad (5)$$

where k denotes the discrete time-step, Δt means the sampling interval, I_{tot}^{pack} denotes the total current of battery pack that indicates discharging when the value is negative, $V_{t,i}^{cell}$ denotes the terminal voltage of i -th cell, $V_{p,i}^{cell}$ denotes the voltage drop across the polarization resistance, $V_{oc,i}^{cell}$ denotes the open-circuit voltage (OCV) which is nonlinear function of the state-of-charge (SoC) denoted by $S_{oc,i}^{cell}$, η_c denotes the Coulomb efficiency that is set to 1, C_a denotes the nominal capacity of cell.

This paper employs fifth-order polynomial to model the nonlinear function $f(S_{oc,i}^{cell}(k))$ as follows:

$$f(S_{oc,i}^{cell}(k)) = \sum_{j=0}^5 (\alpha_{j,i} S_{oc,i}^{cell}(k)^j) \quad (6)$$

where $\alpha_{j,i}$ is the coefficient of i -th cell OCV-SoC polynomial model.

Fig. 3(b) illustrates the battery pack model which can be expressed by:

$$V_{t,i}^{cell}(k) = V_{oc,i}^{cell}(k) + R_{o,i}^{cell}(k) I_{tot}^{pack}(k) + V_{p,i}^{cell}(k) \quad (1)$$

$$V_{p,i}^{cell}(k) = e^{-\Delta t / \tau_i(k)} V_{p,i}^{cell}(k-1) + (1 - e^{-\Delta t / \tau_i(k)}) I_{tot}^{pack}(k) R_{p,i}^{cell}(k) \quad (2)$$

$$\tau_i(k) = R_{p,i}^{cell}(k) C_{p,i}^{cell}(k) \quad (3)$$

$$V_{oc,i}^{cell}(k) = f(S_{oc,i}^{cell}(k)) \quad (4)$$

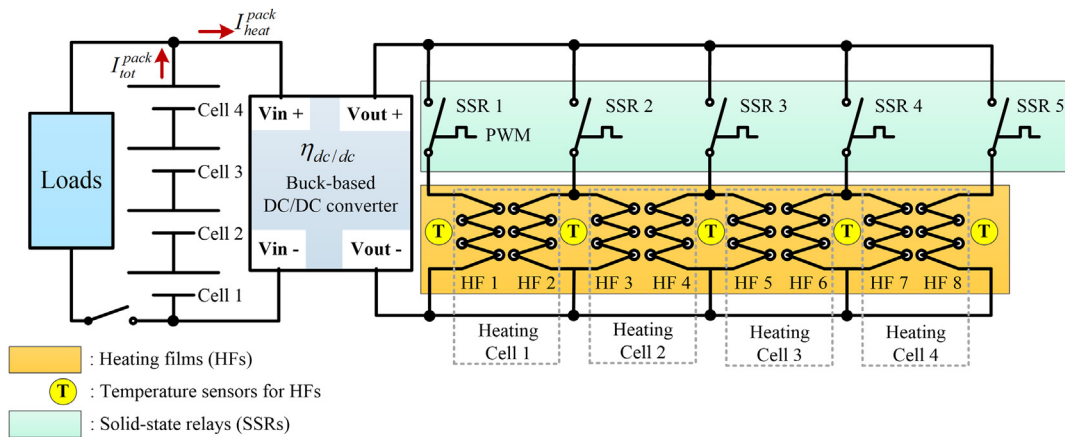


Fig. 2. Self-powered heating circuit for 4S battery pack.

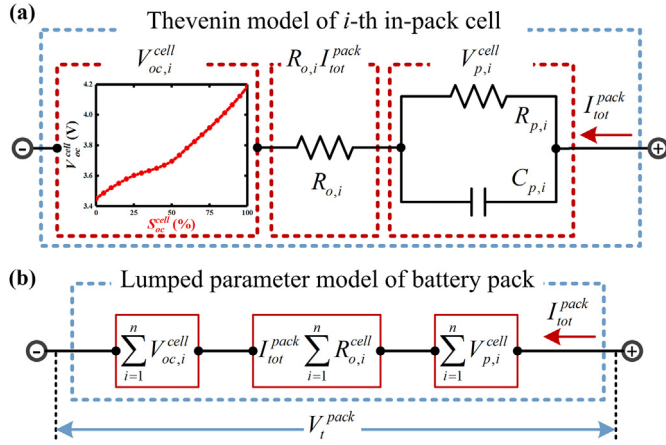


Fig. 3. Electrical equivalent circuit model: (a) Thevenin model of cell. (b) Lumped parameter model of pack.

$$V_t^{pack}(k) = \sum_{i=1}^n V_{oc,i}^{cell}(k) + I_{tot}^{pack}(k) \sum_{i=1}^n R_{o,i}^{cell}(k) + \sum_{i=1}^n V_{p,i}^{cell}(k) \quad (7)$$

$$S_{oc}^{pack}(k) = \begin{cases} \min\{S_{oc,i}^{cell}(k) | i=1,2,\dots,n\}, & \text{if } I_{tot}^{pack}(k) \leq 0 \\ \max\{S_{oc,i}^{cell}(k) | i=1,2,\dots,n\}, & \text{if } I_{tot}^{pack}(k) > 0 \end{cases} \quad (8)$$

where n denotes the total number of in-pack cells, V_t^{pack} and S_{oc}^{pack} denote the terminal voltage and the SoC of battery pack respectively.

To maximize heating speed, all the SSRs mentioned in Fig. 2 are closed during preheating process. Thus, the output power of DC/DC converter $P_{out}^{dc/dc}$ can be calculated by:

$$P_{out}^{dc/dc}(k) = (I_{out}^{dc/dc}(k))^2 R_{hf} \quad (9)$$

where R_{hf} denotes the resistance when all HFs work in parallel, $I_{out}^{dc/dc}$ denotes the output current of Buck circuit-based DC/DC converter.

The heating power of battery pack during preheating, which is equal to the input power of DC/DC converter $P_{in}^{dc/dc}$, can be precisely controlled by regulating $I_{out}^{dc/dc}$ as follows:

$$P_{in}^{dc/dc}(k) = (I_{out}^{dc/dc}(k))^2 R_{hf} / \eta_{dc/dc}(k) \quad (10)$$

$$\eta_{dc/dc}(k) = g_{eff}^{2d}(V_{in}^{dc/dc}(k), I_{in}^{dc/dc}(k)) \quad (11)$$

where g_{eff}^{2d} denotes a 2-dimensional interpolated lookup table which is established based on the efficiency map of DC/DC converter.

Meanwhile, if the permissible heating power of battery pack $P_{heat,max}^{pack}$ is known, the target value of $I_{out}^{dc/dc}$ can be deduced from Eq [10], and [11] by setting $P_{in}^{dc/dc}(k) = P_{heat,max}^{pack}(k)$:

$$I_{out}^{dc/dc}(k) = \sqrt{P_{heat,max}^{pack}(k) g_{eff}^{2d}(V_t^{pack}(k), I_{heat}^{pack}(k)) / R_{hf}} \quad (12)$$

Notice that Eq [12], plays the key role in heating power constraint, and the value of $P_{heat,max}^{pack}$ is specified in Section 3.2.

2.3. Thermal dynamic modeling of battery pack

To facilitate online identification of battery thermal characteristic parameters, the thermal dynamic of battery pack can be modeled by utilizing equivalent circuit model [36] in Fig. 4(a). To improve energy-efficiency of heating, the battery pack is put into a battery box pasted with heat insulation material for heat preservation. T^{box} is the temperature (K) in battery box and is also the ambient temperature of battery pack. $R_{h,i}^{cell} |_{i=1,2,3,4}$ represents the thermal resistance (K/W) between individual cell and surroundings. $C_{h,i}^{cell} |_{i=1,2,3,4}$ represents the thermal capacity (J/K) of individual cell. $R_{h,i}^a, R_{h,i}^b, R_{h,i}^c$ are the thermal resistances (K/W) between adjacent in-pack cells. $Q_{ins,i}^{cell} |_{i=1,2,3,4}$ and $Q_{os,i}^{cell} |_{i=1,2,3,4}$ represent the internal heat generation rate and the external heat generation rate (W) of individual cell respectively. The thermal dynamic of battery pack can be expressed by:

$$\sum_{i=1}^4 C_{h,i}^{cell} \frac{dT_i^{cell}}{dt} = \sum_{i=1}^4 (Q_{ins,i}^{cell} + Q_{os,i}^{cell}) + \sum_{i=1}^4 \frac{T^{box} - T_i^{cell}}{R_{h,i}^{cell}} \quad (13)$$

Considering that the external heat generation rates ($Q_{os,i}^{cell} |_{i=1,2,3,4}$) of all the cells are different caused by the above-mentioned heating structure and cannot be measured directly, the thermal model of battery pack needs to be simplified as shown in Fig. 4(b). Since the thermal capacity and the thermal resistance are slowly time-varying parameters, Eq [13], can be rewritten as follows after discretization:

$$C_h^{pack}(k) \frac{T_{avg}^{pack}(k) - T_{avg}^{pack}(k-1)}{\Delta t} + \frac{T^{box}(k-1) - T_{avg}^{pack}(k-1)}{R_h^{pack}(k)} = Q_{tot}^{pack}(k-1) \quad (14)$$

$$T_{avg}^{pack}(k) = \frac{1}{n} \sum_{i=1}^4 T_i^{cell}(k) \quad (15)$$

$$Q_{tot}^{pack}(k) = Q_{ins}^{pack}(k) + Q_{os}^{pack}(k) \quad (16)$$

where C_h^{pack} denotes the lumped thermal capacity of battery pack, R_h^{pack} denotes the lumped thermal resistance between the whole battery pack and the surroundings, T_{avg}^{pack} is the average value of all the cell temperatures, $n = 4$ is the total number of cells, Q_{ins}^{pack} and Q_{os}^{pack} denote the internal heat generation rate and the external one of battery pack respectively and can be calculated by:

$$Q_{ins}^{pack}(k) = (V_{oc}^{pack}(k) - V_t^{pack}(k)) | I_{tot}^{pack}(k) | \quad (17)$$

$$Q_{os}^{pack}(k) = V_t^{pack}(k) | I_{heat}^{pack}(k) | \eta_{dc/dc}(k) \quad (18)$$

The reversible entropic heat of internal heat generation is ignored in Eq [17], because it is fairly small compared with the irreversible heat [37]. Q_{os}^{pack} is equal to the output power of DC/DC converter $P_{out}^{dc/dc}$.

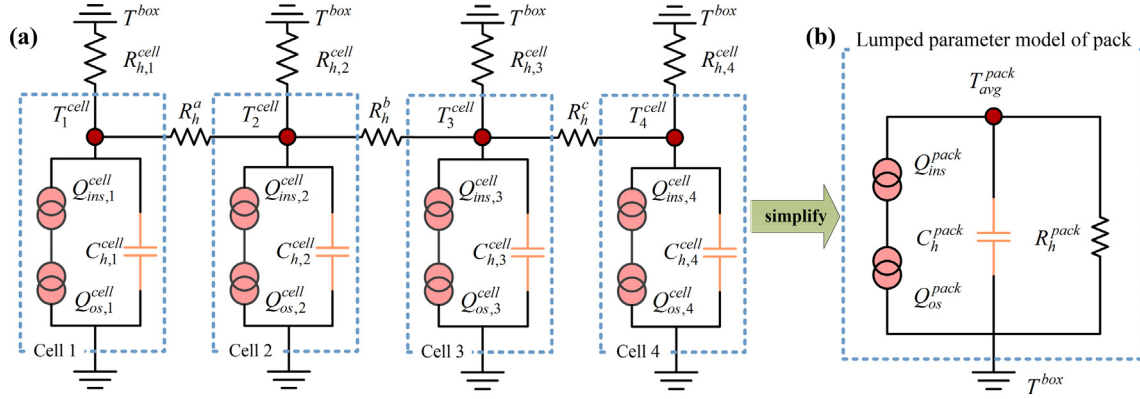


Fig. 4. Thermal equivalent circuit model of battery pack: (a) The original form. (b) The simplified form.

2.4. Identification of model parameters

To achieve the extraction of above-mentioned model parameters, Eq [1], and Eq [14], need to be converted into multiple regression forms as follows:

$$E_i(k) = [E_i(k-1), I_{tot}^{pack}(k), I_{tot}^{pack}(k-1)] \cdot [\theta_{1,i}^e(k), \theta_{2,i}^e(k), \theta_{3,i}^e(k)]^T \quad (19)$$

$$D(k) = [D(k-1), Q_{tot}^{pack}(k-1), T^{box}(k-1)] \cdot [\theta_1^t(k), \theta_2^t(k), \theta_3^t(k)]^T \quad (20)$$

where for the electrical equivalent circuit model (EECM):

$$E_i(k) = V_{t,i}^{cell}(k) - V_{oc,i}^{cell}(k) \quad (21)$$

$$\theta_{1,i}^e(k) = \frac{\Delta t_e - 2R_{p,i}^{cell}(k)C_{p,i}^{cell}(k)}{\Delta t_e + 2R_{p,i}^{cell}(k)C_{p,i}^{cell}(k)} \quad (22a)$$

$$\theta_{2,i}^e(k) = \frac{\Delta t_e (R_{o,i}^{cell}(k) + R_{p,i}^{cell}(k)) + 2R_{p,i}^{cell}(k)C_{p,i}^{cell}(k)}{\Delta t_e + 2R_{p,i}^{cell}(k)C_{p,i}^{cell}(k)} \quad (22b)$$

$$\theta_{3,i}^e(k) = \frac{\Delta t_e (R_{o,i}^{cell}(k) + R_{p,i}^{cell}(k)) - 2R_{p,i}^{cell}(k)C_{p,i}^{cell}(k)}{\Delta t_e + 2R_{p,i}^{cell}(k)C_{p,i}^{cell}(k)} \quad (22c)$$

where for the thermal equivalent circuit model (TECM):

$$D(k) = T_{avg}^{pack}(k) \quad (23)$$

$$\theta_1^t(k) = 1 - \frac{\Delta t_t}{R_h^{pack}(k)C_h^{pack}(k)} \quad (24a)$$

$$\theta_2^t(k) = \frac{\Delta t_t}{C_h^{pack}(k)} \quad (24b)$$

$$\theta_3^t(k) = \frac{\Delta t_t}{R_h^{pack}(k)C_h^{pack}(k)} \quad (24c)$$

Based on Eq [19], and Eq. [20], the parameter vectors, denoted by $\Theta_i^e(k) = [\theta_{1,i}^e(k), \theta_{2,i}^e(k), \theta_{3,i}^e(k)]^T$ and $\Theta^t(k) = [\theta_1^t(k), \theta_2^t(k), \theta_3^t(k)]^T$,

can be extracted online by using the recursive least squares (RLS) algorithm. The discrete time-steps denoted by Δt_e and Δt_t are set to 0.1 s and 1 s respectively. The iterative calculation of RLS consists of following five steps:

1. Update the input vector $\Phi(k)$ and the output vector $y_{rls}(k)$:

$$\begin{cases} \Phi_i^e(k) = [E_i(k-1), I_{tot}^{pack}(k), I_{tot}^{pack}(k-1)] \\ y_{rls,i}^e(k) = E_i(k) \end{cases} \quad (25)$$

$$\begin{cases} \Phi^t(k) = [D(k-1), Q_{tot}^{pack}(k-1), T^{box}(k-1)] \\ y_{rls}^t(k) = T_{avg}^{pack}(k) \end{cases} \quad (26)$$

2. Compute the gain matrix $K_{rls}(k)$ and update the covariance matrix $P_{rls}(k)$:

$$K_{rls}(k) = P_{rls}(k-1)\Phi(k)^T [\lambda_{rls}(k-1) + \Phi(k)P_{rls}(k-1)\Phi(k)^T]^{-1} \quad (27)$$

$$P_{rls}(k) = \lambda_{rls}(k-1)^{-1} [P_{rls}(k-1) - K_{rls}(k)\Phi(k)P_{rls}(k-1)] \quad (28)$$

where λ_{rls} denotes the forgetting factor, and $\lambda_{rls} \in [0.8, 1]$.

3. Estimate the parameter vector $\Theta(k)$:

$$\Theta(k) = \Theta(k-1) + K_{rls}(k)(y_{rls}(k) - \Phi(k)\Theta(k-1)) \quad (29)$$

4. Solve the model parameters of EECM and TECM from $\Theta(k)$ according to Eq.(22a) ~ (22c) and Eq.(24a) ~ (24c) respectively.

5. Increase the discrete time-step k by 1 and return to step 1 for next iteration.

Since the parameters of EECM are strongly time-varying when cell temperature changes fast, the adaptive adjustment of λ_{rls} is employed in the online identification of EECM parameters:

$$\beta_{rls}(k) = 2^{\rho(y_{rls}(k) - \Phi(k)\Theta(k-1))^2} \quad (30a)$$

$$\lambda_{rls}(k) = \lambda_{\min} + (1 - \lambda_{\min})^{\beta_{rls}(k)} \quad (30b)$$

where $\lambda_{\min}$ denotes the minimum value of λ_{rls} and is set to 0.8. Notice that the RLS with a variable forgetting factor is called as the adaptive recursive least squares (ARLS). In addition, λ_{rls} is constant

Table 2
Procedure of battery pack SoP prediction.

1. Setting Define the lengths of future steps L : Short-term length: $L_s = 100$, Long-term length: $L_l = 300$.	
2. SoP prediction for in-pack cell	
2.1 Current constraint ^a : $\hat{P}_{cc}^{cell} = V_{t,min}^{cell} I_{cc,max}^{cell}$	[31]
2.2 SoC constraint ^b : $\hat{P}_{soc,i}^{cell,L}(k) = V_{t,min}^{cell} \frac{\hat{S}_{oc,i}^{cell}(k) - S_{oc,min}^{cell}}{\eta_c L \Delta t_e / 3600 C_a}$	[32]
2.3 Direct-current resistance (DCR) constraint ^c : $\hat{P}_{dcr}^{cell}(k) = V_{t,min}^{cell} \frac{V_{oc}^{cell}(k) - V_{t,min}^{cell}}{R_{dc,i}^{cell}(k)}$	[33]
2.4 Dynamic voltage constraint:	[34]
$\hat{P}_{volt,i}^{cell,L}(k) = V_{t,min}^{cell} \frac{V_{oc,i}^{cell}(k) + (e^{-\Delta t_e / \tau_i(k)})^L V_{p,i}^{cell}(k) - V_{t,min}^{cell}}{\eta_c L \Delta t_e \left. \frac{dV_{oc,i}^{cell}}{dS_{oc,i}^{cell}} \right _{S_{oc,i}^{cell} = \hat{S}_{oc,i}^{cell}(k)} + R_{o,i}^{cell}(k) + (1 - e^{-\Delta t_e / \tau_i(k)}) R_{p,i}^{cell}(k) \sum_{j=0}^{L-1} (e^{-\Delta t_e / \tau_i(k)})^{L-1-j}}$	
2.5 Calculate the SoP based on Eq [31], ~ [34] through setting $L = L_s$ and $L = L_l$ respectively.	
2.6 Filter the SoP of in-pack cell with multiple constraints: $\hat{P}_{max,i}^{cell,Ls}(k) = \min\{\hat{P}_{cc}^{cell,Ls}(k), \hat{P}_{soc,i}^{cell,Ls}(k), \hat{P}_{dcr}^{cell,Ls}(k), \hat{P}_{volt,i}^{cell,Ls}(k)\}$, $\hat{P}_{max,i}^{cell,Ll}(k) = \min\{\hat{P}_{cc}^{cell,Ll}(k), \hat{P}_{soc,i}^{cell,Ll}(k), \hat{P}_{dcr}^{cell,Ll}(k), \hat{P}_{volt,i}^{cell,Ll}(k)\}$	
3. SoP prediction for battery pack Predict the SoP of battery pack ^d :	
$\hat{P}_{max}^{pack,Ls}(k) = n \cdot \min\{\hat{P}_{max,i}^{cell,Ls}(k)\}_{i=1,2,\dots,n}$, $\hat{P}_{max}^{pack,Ll}(k) = n \cdot \min\{\hat{P}_{max,i}^{cell,Ll}(k)\}_{i=1,2,\dots,n}$	
^a is set to 6 C-rate (60 A), $V_{t,min}^{cell}$ is set to 3 V.	
^b denotes the minimum SoC of in-pack cell and is set to 0.	
^c is calculated based on a lookup table $R_{dc,i}^{cell}(k) = g_{rdc}^{2d}(T_i^{cell}(k), S_{oc,i}^{cell}(k))$ which is established in Ref. [35].	
^d $\hat{P}_{max}^{pack,Ls}$ and $\hat{P}_{max}^{pack,Ll}$ denote the short-term SoP and the long-term SoP of battery pack respectively.	

in the RLS employed for the online identification of TECM parameters.

3. Description of heating strategy

This section proposes a model prediction-based heating strategy which includes two processes [1]: The preheating is designed to rapidly restore the power capability of battery pack through warming it up from -40°C [2]. The temperature holding is designed to control the cell temperatures adaptively during the normal operation of the preheated battery pack.

3.1. Co-estimation of multiple battery states

Multiple parameters and states of battery pack as the essential inputs of strategy need to be co-estimated based on the established models in real-time. To reduce the computational complexity of online iterative calculation, this paper employs the extended Kalman filter (EKF) which cooperates with the ARLS to update the parameters and the SoC of battery pack based on EECM. The well-known EKF algorithm is listed in TABLE A1 in Appendix A.

The state-of-power (SoP) is defined to describe the maximum power capability of battery pack. To prevent lithium-ion deposition, the process of charging battery pack at low temperatures is not allowed in this paper. Therefore, only the SoP during discharging process needs to be predicted, and it is defined as the maximum power that can be utilized to discharge battery pack with a constant current profile in a specified period L . Specifically, dual time-scales denoted by L_s and L_l are employed in SoP prediction to capture the short-term and long-term power capability of battery pack respectively. The details of calculating battery pack SoP with multiple constraints are listed in Table 2.

Based on Eq [14], ~ [18], the heating power of battery pack to stabilize its temperature at a setting value can be calculated by:

$$P_{heat,tar}^{pack}(k) = \frac{1}{\eta_{dc/dc}(k)} \left(\frac{T_{tar}^{pack}(k) - T^{box}(k)}{R_h^{pack}(k)} - \sum_{\tau=k-L_l+1}^k Q_{ins}^{pack}(\tau) \right) \quad (35)$$

where $P_{heat,tar}^{pack}$ denotes the target of external heating power utilized

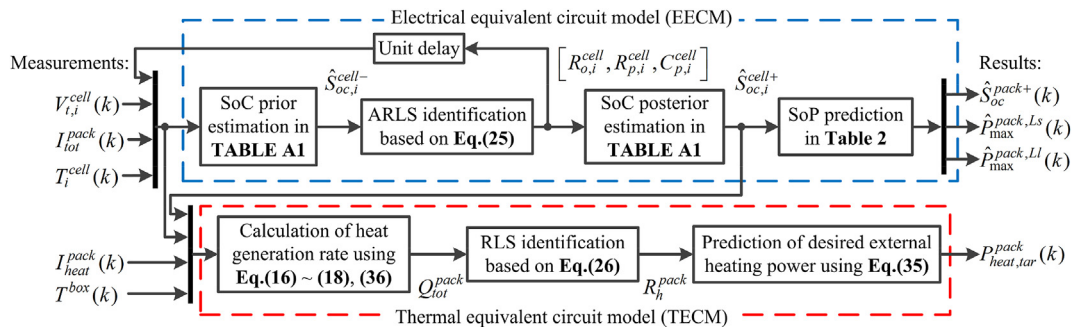


Fig. 5. Flowchart of the model-based battery states co-estimation.

to hold battery pack temperature at a target value denoted by T_{tar}^{pack} . Considering the drastic change of battery current, a moving average filter with a window length of L_l is used to evaluate the internal heat generation rate.

To estimate the battery states online, an adaptive co-estimation framework is developed in Fig. 5. Notice that the moving average value of the total heat generation rate is inputted into the RLS of

thermal parameter identification for better algorithm performance, which can be expressed by:

$$Q_{tot}^{pack}(k) = \sum_{\tau=k-L_s+1}^k (Q_{ins}^{pack}(\tau) + Q_{os}^{pack}(\tau)) \quad (36)$$

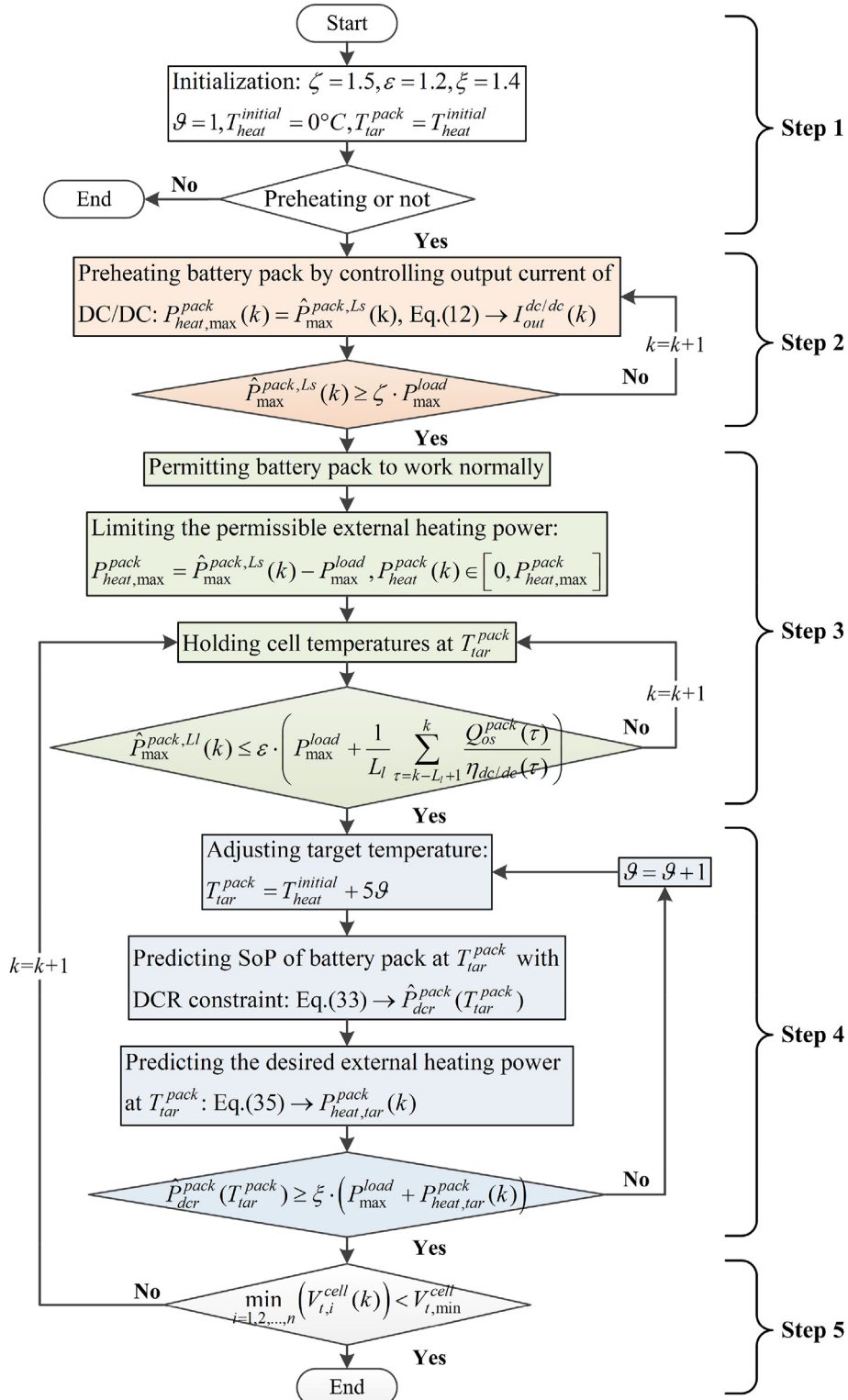


Fig. 6. Framework of the proposed heating strategy.

3.2. Framework of model prediction-based heating strategy

The strategy is designed to maximize heating speed in the preheating process and to raise temperature stepwise in the temperature holding process. The framework of the heating strategy is illustrated in Fig. 6. The model prediction results are utilized as the inputs of the strategy and they play the key role in the process switch. The procedures of implementing the strategy are explained in detail as follows:

Step 1: Initialization. Set the initial parameters of the strategy for the process switch, which are denoted by $[\zeta, \varepsilon, \xi]$, and these parameters are specified depending on the practical loading condition and the size of battery pack. Judge whether preheat battery pack or not. If yes, execute Step 2 for preheating. If not, allow battery pack to work normally.

Step 2: Preheating. Battery pack is not permitted to output power for external loads except for the heating films (HF), and the external heating power outputted from battery pack is regulated by controlling the output current of DC/DC converter [1]: Set the permissible heating power as $P_{heat,max}^{pack} = \hat{P}_{max}^{pack,LS}$ to maximize preheating speed and calculate the controlled target $I_{out}^{dc/dc}(k)$ based on Eq [12]. Then, control the DC/DC converter to output the mixed current combined from pulse current and continuous current at amplitude of $I_{out}^{dc/dc}(k)$ [2]. Keep preheating the battery pack and judge whether or not the short-term SoP $\hat{P}_{max}^{pack,LS}$ is greater than $\zeta \cdot P_{load,max}^{load}$. If yes, execute Step 3 for temperature holding. If not, continue to preheat battery pack with the power of $\hat{P}_{max}^{pack,LS}$. The profile of the mixed current is illustrated in Fig. 7. The reason of generating the mixed current for preheating is to improve the convergence of the core algorithms mentioned in Fig. 5. The value of $P_{load,max}^{load}$ will be specified in Section 4.2 according to the employed dynamic loading conditions.

Step 3: Holding temperature. Battery pack is permitted to power the external loads (e.g., electric motor of EVs). Restrict the permissible external heating power to $P_{heat,max}^{pack} = \hat{P}_{max}^{pack,LS} - P_{load,max}^{load}$ for ensuring that the battery pack can afford $P_{load,max}^{load}$. Keep raising the cell

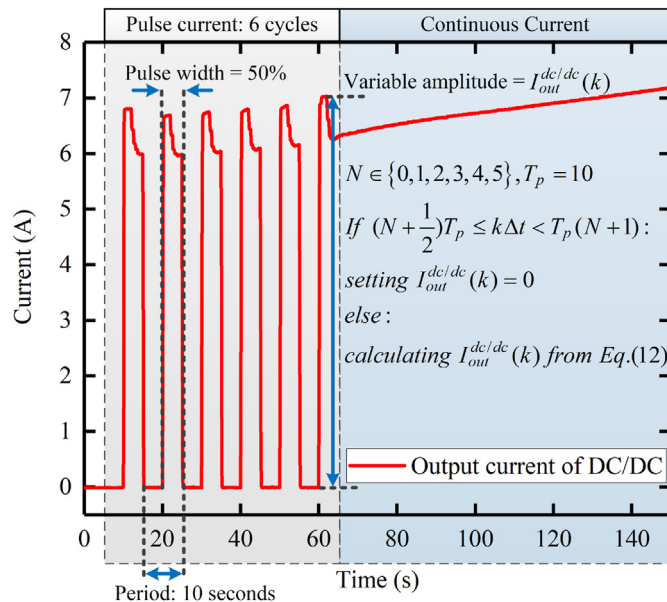


Fig. 7. Profile of the output current of DC/DC converter during preheating.

temperatures gently by stabilizing the HF's temperatures at T_{tar}^{pack} , and judge whether or not the criterion described in Eq [37]. is satisfied. If yes, execute Step 4 for boosting power. If not, continue to hold the HF's temperatures at T_{tar}^{pack} for saving energy.

$$\hat{P}_{max}^{pack,LI}(k) \leq \varepsilon \cdot \left(P_{load,max}^{load} + \frac{1}{L_l} \sum_{\tau=k-L_l+1}^k \frac{Q_{os}^{pack}(\tau)}{\eta_{dc/dc}(\tau)} \right) \quad (37)$$

Step 4: Adjusting temperature. Considering that the thermal capacity of in-pack cell is much greater than that of HF, continuously changing the value of the controlled temperature T_{tar}^{pack} will enlarge the undesired temperature difference between the HF and the in-pack cell during the temperature holding. To improve the accuracy of temperature control, the strategy is designed to choose 5 °C as the step and increase T_{tar}^{pack} stepwise by setting $T_{tar}^{pack} = T_{heat}^{initial} + 5\vartheta$. Predict the SoP of battery pack at the updated target T_{tar}^{pack} based on Eq. [33], meanwhile, predict the desired external heating power at T_{tar}^{pack} based on Eq [35]. Then, judge whether or not the criterion described in Eq. (38) is satisfied. If yes, execute Step 5 for checking cell voltage. If not, set $\vartheta = \vartheta + 1$ to further increase T_{tar}^{pack} .

$$\hat{P}_{dcr}^{pack}(T_{tar}^{pack}) \geq \xi \cdot (P_{load,max}^{load} + P_{heat,max}^{pack}(k)) \quad (38)$$

Step 5: Checking voltage. Judge whether or not the measured terminal voltage of in-pack cell is less than the lower cut-off voltage. If yes, end the operation of battery pack for protection. If not, return to Step 3 for holding the HF's temperatures at the updated target T_{tar}^{pack} .

4. Experiments

A series of battery-in-the-loop tests is conducted based on a real-time experimental platform in this section. The battery pack mentioned in Section 2.1 is selected as heating object for the tests.

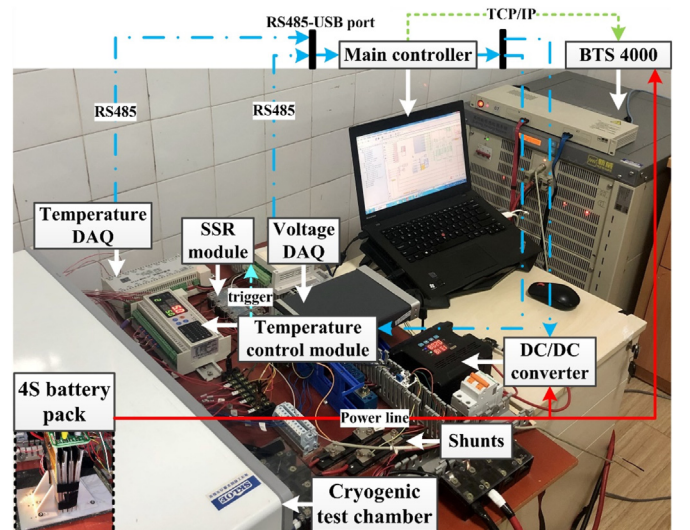


Fig. 8. Configuration of the battery-in-the-loop test platform.

Table 3
Specifications of the devices.

Device	Parameter
Buck DC/DC converter	960 W, 16 A, $\pm 0.05\%$
NEWARE BTS 4000	30 V, 100 A, $\pm 0.05\%$
Voltage DAQ	± 10 V, $\pm 0.05\%$
Current shunt	50 A, 50 mV, $\pm 0.5\%$
Temperature DAQ	-200 – 600 °C, $\pm 0.05\%$
Cryogenic test chamber	-65 – 25 °C, $\pm 1\%$

4.1. Experimental platform

The experimental platform is established as shown in Fig. 8, and the specifications of the employed devices are described in Table 3. A personal computer is utilized to act as the main controller. The models, co-estimation algorithms and strategy are embedded into the main controller through MATLAB/Simulink software. The updated measurements including voltages, currents and temperatures are inputted into the main controller at the frequency of 10 Hz, and meanwhile the main controller runs the embedded algorithms and strategy at the same frequency. The main controller sends the control signals of heating power and heating temperature to the DC/DC converter and the temperature control module (TCM) respectively at the frequency of 2 Hz. The TCM is employed to achieve the heating temperature control based on the embedded PID algorithm, and the SSR module is controlled by the trigger signal generated from the TCM. A battery test system (BTS) is employed in the charging and dynamic loading of battery pack. A cryogenic test chamber is utilized to create the extremely cold environment.

4.2. Test schedule

The procedures of the tests are summarized in Fig. 9. The temperature in the test chamber is set to -40 °C. As listed in Table 4, two test groups composed of eight tests, denoted by Test I ~ Test VIII, are set up for verification. Before each test, the battery pack is fully charged by using BTS 4000 at 25 °C, and then the battery pack is put into a battery box inside the test chamber and allowed to rest at -40 °C for 6 h. At the beginning of the test, the temperature of battery pack and the temperature in the battery box are both

stabilized at -40 °C. The same preheating process is performed according to the proposed strategy in the eight tests, and then two different test cycles are loaded on battery pack during the holding process. The employed test cycles are the dynamic stress test (DST) and the federal urban driving schedule (FUDS), and all the charging steps of them are deleted to avoid lithium-ion deposition. The maximum power loading in DST and FUDS is set to the rated power of the tested battery pack when discharging with 2C-rate current (20 A). Hence, the maximum power demand of external loads $P_{\max}^{\text{load}}$ in Fig. 6 is set to 240 W. In holding process, Test I and Test V follow the strategy to execute the stepwise temperature control with the variable target of $T_{\text{tar}}^{\text{pack}}$. For comparison, the constant temperature control is performed in the holding process of Test II ~ IV and Test VI ~ VIII, and the target temperatures of these compared tests are fixed at 0 °C, 5 °C and 10 °C, as shown in Table 4. All the tests follow the same procedures in Fig. 9.

4.3 Results and discussion.

To clearly reveal a series of experimental results in a short place, only the details of the two tests (Test I and Test V) that execute the whole strategy are illustrated in Fig. 10 and Fig. 11 respectively.

4.3.1 Results of Test I in Test group 1.

Fig. 10 (a) plots the currents outputted from battery pack. During the preheating, the value of the total current is equal to that of the heating current, which indicates that the permissible power of battery pack is usefully employed in heating. The preheating is finished at 380 s and at the same instant the battery pack starts to load the modified DST cycles. The temperature measurements are plotted in Fig. 10 (b). The average temperature of pack rises from -41.2 °C to -14.3 °C in the preheating, and then the pack temperature keeps rising till it stabilizes at the target $T_{\text{tar}}^{\text{pack}}$ in the temperature holding. The temperature $T_{\text{tar}}^{\text{pack}}$ is controlled to rise from 0 °C to 15 °C during the whole test. Notice that although $T_{\text{tar}}^{\text{pack}}$ increases to 15 °C at 6814 s, the battery pack cannot further provide enough power for heating due to low SoP, which consequently sends pack temperature plummeting. The results of co-estimation and prediction based on the EECM are illustrated in Fig. 10 (c) ~ (f). The estimated terminal voltage of battery pack is compared with the measurements in Fig. 10 (c). The estimation error of pack voltage is plotted in Fig. 10 (d), and the error is relatively larger in the preheating caused by the dramatic temperature

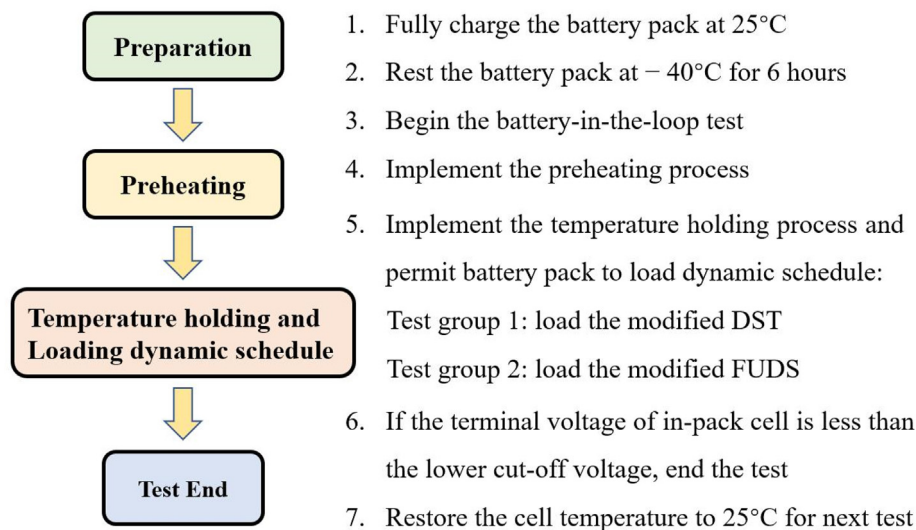
**Fig. 9.** Details of implementing the battery-in-the-loop test.

Table 4
Cases of the battery-in-the-loop test.

Identical preheating	Test group 1				Test group 2			
	Test I	Test II	Test III	Test IV	Test V	Test VI	Test VII	Test VIII
Dynamic Loads	DST	DST	DST	DST	FUDS	FUDS	FUDS	FUDS
Target temperatures	T_{tar}^{pack}	0 °C	5 °C	10 °C	T_{tar}^{pack}	0 °C	5 °C	10 °C

rise. The comparison of the estimated SoC with the reference SoC is plotted in Fig. 10 (e), and the current integral method is utilized to calculate the reference SoC based on the measured current. The statistic results of co-estimation, in terms of the root mean square error (RMSE) and the mean absolute error (MAE), are listed in

Table 5. The profiles of the predicted SoP and the measured total output power of battery pack are plotted in Fig. 10 (f). The prediction results of the short-term SoP and the long-term SoP are getting closer after raising the pack temperature to above 0 °C. The total output power is very close to the predicted 10-s short-term SoP in

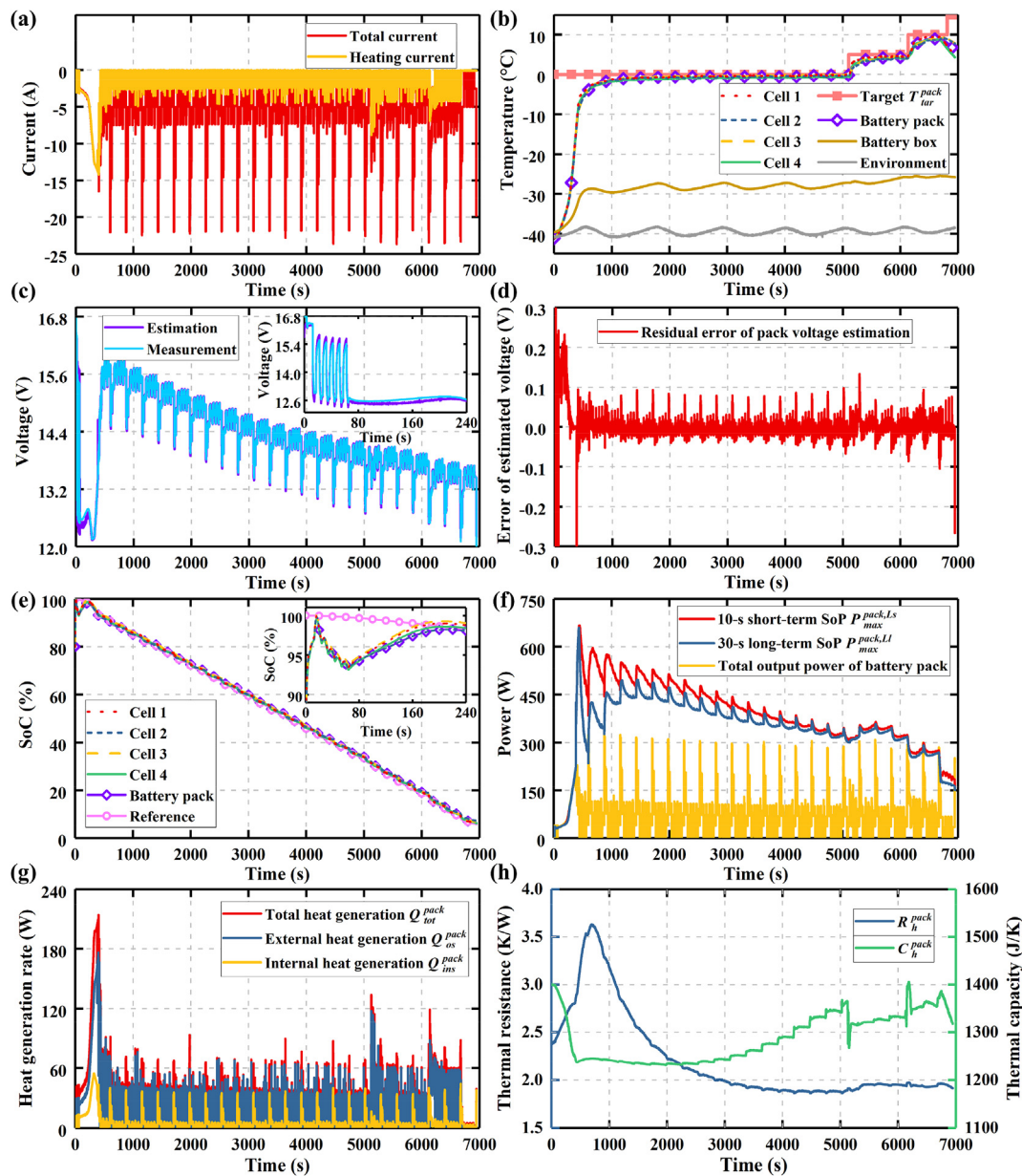


Fig. 10. Results of Test I: (a) Profiles of the currents. (b) Profiles of the measured temperatures. (c) Profiles of the measured voltage and the estimated voltage. (d) Profiles of voltage estimation error. (e) Profiles of the estimated SoC and the reference SoC. (f) Profiles of the predicted SoP and the total output power. (g) Profiles of the heat generation rates. (h) Profiles of the identified thermal parameters.

the preheating, which indicates that the developed model and control of the DC/DC converter are accurate. The calculation results based on the TECM are illustrated in Fig. 10 (g) ~ (h). The internal heat generation rate is relatively larger in the preheating due to the larger DCR caused by low temperatures, which helps to efficiently preheat battery pack by using the limited battery power.

4.3.2 Results of Test V in Test group 2.

Fig. 11 (a) provides the comparison of the total current with the heating current. In the temperature holding, the average values of the total current and the heating current are 4.8 A and 1.0 A respectively, which indicates that the heating current only accounts for a small part of the total current when battery pack loads the modified FUDS cycles. Fig. 11 (b) shows the measurements of various temperatures. This test takes 371 s to finish the preheating, in which the pack temperature varies from -41.4°C to -14.0°C by consuming 6.1% of nominal capacity. Fig. 11 (c) reveals the

estimation results and the measurements of battery pack voltage, and the error of the estimated voltage based on EECM is illustrated in Fig. 11 (d). Besides, Fig. 11 (e) plots the profiles of SoC estimation results, and this paper sets the initial value of SoC estimation as 80%. The statistic results of both the voltage and SoC estimation are listed in Table 6. Fig. 11 (f) plots the prediction results of SoP and the measurements of the total output power. According to the electrical characteristic of LIBs, the available power capability of battery pack can be indirectly reflected in its terminal voltage. From Fig. 11(c) and (f), it can be seen that the pack voltage is very close to the lower cut-off voltage when the total output power is greater than the 10-s short-term SoP, which indicates that the predicted 10-s short-term SoP can well describe instantaneous power. Fig. 11 (g) plots the calculation results of various heat generation rates based on Eq [16]. ~ [18]. The average value of external heat generation rate is 3 times greater than that of internal heat generation rate, which

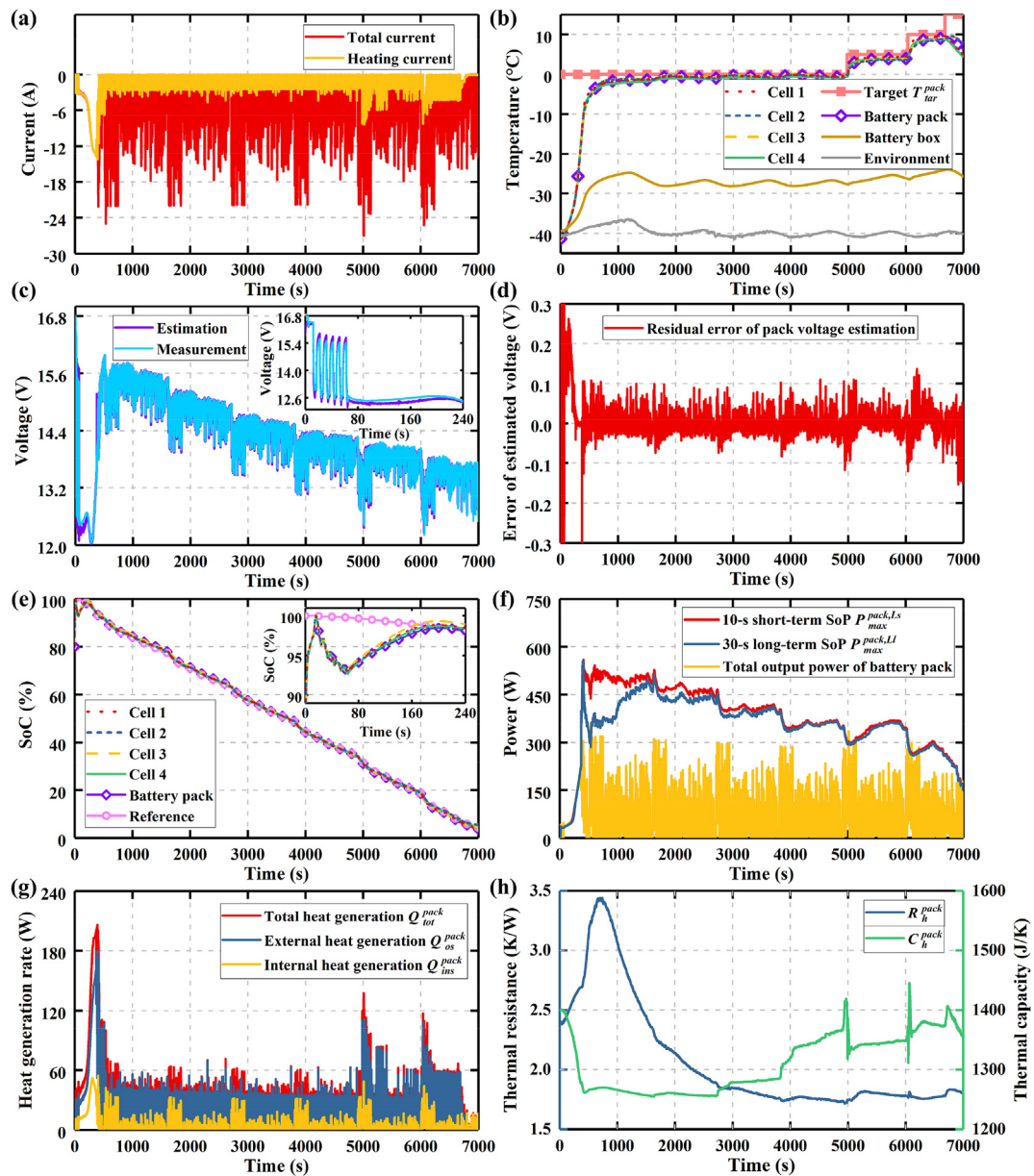


Fig. 11. Results of Test V: (a) Profiles of the currents. (b) Profiles of the measured temperatures. (c) Profiles of the measured voltage and the estimated voltage. (d) Profiles of voltage estimation error. (e) Profiles of the estimated SoC and the reference SoC. (f) Profiles of the predicted SoP and the total output power. (g) Profiles of the heat generation rates. (h) Profiles of the identified thermal parameters.

Table 5

Statistic results of co-estimation for battery pack in Test I.

Test I	Terminal voltage (mV)	SoC (%)
RMSE	40.97	0.77
MAE	18.88	0.46

Table 6

Statistic results of co-estimation for battery pack in Test V.

Test V	Terminal voltage (mV)	SoC (%)
RMSE	45.83	0.78
MAE	23.24	0.41

confirms that the proposed external heating structure plays a significant role in holding the pack temperature. Fig. 11 (h) plots the identification results of TECM's parameters. The thermal resistance of battery pack denoted by R_h^{pack} is a crucial parameter in the prediction of the desired external heating power as expressed in Eq [35].

4.3.3 Evaluation of strategy performance.

The consumption of battery pack capacity in all the tests is listed in Table 7. The relationship of the capacities reported in Table 7 can be expressed by:

$$C_a^{pack} = C_{a,tot}^{pack} + C_{a,f}^{pack} \quad (39a)$$

$$C_{a,tot}^{pack} = C_{a,w}^{pack} + C_{a,h}^{pack} \quad (39b)$$

where C_a^{pack} denotes the nominal capacity of the battery pack and is 10 Ah, $C_{a,tot}^{pack}$ denotes the total discharged capacity, $C_{a,w}^{pack}$ denotes the working capacity consumed by loading the dynamic cycles, $C_{a,h}^{pack}$ denotes the capacity consumed by powering the external heating in both the preheating and the temperature holding, $C_{a,f}^{pack}$ denotes the frozen capacity.

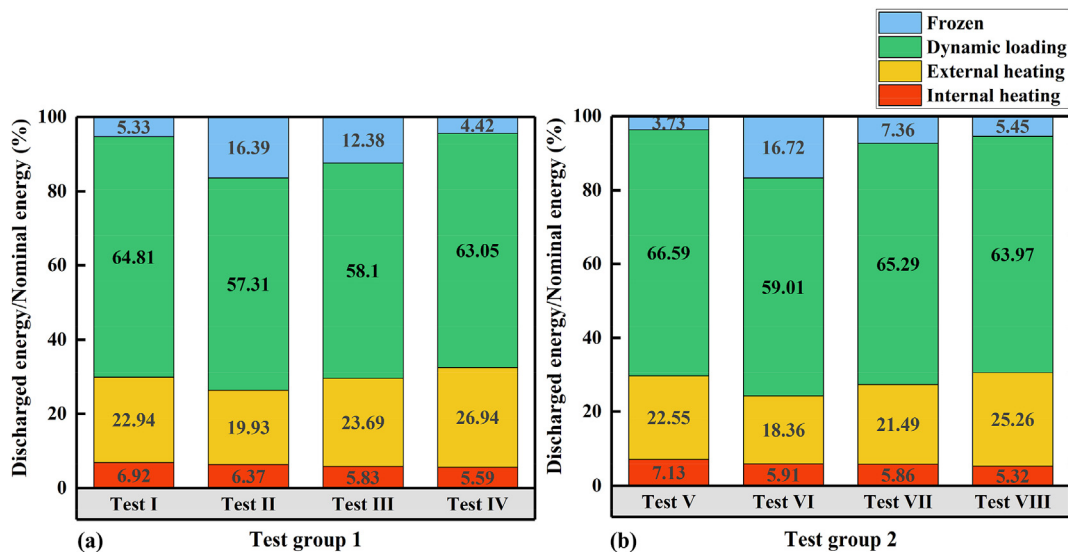
Test I and Test V show the largest working capacity compared with the other tests in their respective test groups, which indicates that the stepwise temperature holding in the strategy is useful for maximizing the available working capacity. Furthermore, although keeping the pack temperature at a higher level in the temperature holding process can increase the total discharged capacity, it will also increase the heating consumption, so that the working capacity will not always increase with the rising pack temperature.

The detailed consumption of battery pack energy in all the tests is illustrated in Fig. 12. The nominal energy is defined as the discharged energy when battery totally releases the nominal capacity with 0.5 C-rate constant current at 25 °C, and the nominal energy of the selected battery pack is 150 Wh. In Fig. 12, the green bar denotes the proportion of the battery energy consumed by loading the dynamic cycles after preheating, and the strategy aims to increase this amount of energy when battery pack operates at − 40 °C. The yellow bar denotes the proportion of the battery energy consumed by the HF's for external heat generation. The red bar denotes the proportion of the battery energy consumed by the Joule effect occurring on the internal resistance of battery pack. Notice that increasing the controlled temperature in the temperature holding process can reduce the frozen energy, but it will also increase the energy consumption in external heating. Meanwhile, the internal

Table 7

Discharged capacity of battery pack in different tests.

Test cases		Test group 1: load DST				Test group 2: load FUDS			
		Test I	Test II	Test III	Test IV	Test V	Test VI	Test VII	Test VIII
Test duration (s)		6947	6123	6419	6680	7012	6023	6454	6198
Total	$C_{a,tot}^{pack}$ (Ah)	9.42	8.24	8.65	9.52	9.59	8.21	9.19	9.40
Working	$C_{a,w}^{pack}$ (Ah)	6.93	6.09	6.13	6.65	7.13	6.19	6.82	6.57
Heating	$C_{a,h}^{pack}$ (Ah)	2.49	2.15	2.52	2.87	2.46	2.02	2.37	2.83
Frozen	$C_{a,f}^{pack}$ (Ah)	0.58	1.76	1.35	0.48	0.41	1.79	0.81	0.60

**Fig. 12.** Comparison of energy consumptions in different tests: (a) Test group 1. (b) Test group 2.

resistance of in-pack cell will decrease with the rising temperature, which will consequently lead to the reduction of the energy consumption in internal heating. Considering that the energy-efficiency of internal heating is higher than that of external heating, the available energy of battery pack supplied to the loads can be increased by properly improving the proportion of internal heating in the total energy consumption of heating. The strategy is therefore designed to raise the controlled temperature stepwise. As shown in Fig. 12, Test I and Test V present the largest energy supplied to the loads in their respective test groups, which indicates that the strategy can efficiently utilize the restored energy of battery pack after preheating.

5. Conclusions

This paper has proposed a novel battery self-powered heating strategy, which can make the commonly used commercial LIBs suitable for operating at the ECT of -40°C . The strategy contains two processes, namely the preheating and the temperature holding. The features of the strategy are summarized as follows:

- (1) An external heating structure and a self-powered heating circuit were developed for the series-connected battery pack to support the implementation of the strategy, which allows the battery pack to supply the heating power while also supplying power to the other external load. Multiple parameters and states of the battery pack, such as SoC and SoP, were co-estimated based on battery models in real-time to provide the essential inputs of the strategy.
- (2) Heating power was controlled in the preheating to maximize heating speed and meanwhile to prevent battery pack from being over-discharged. Heating temperature was controlled in the temperature holding to improve energy-efficiency.
- (3) The strategy was designed to switch from the preheating to the temperature holding when the predicted SoP of the preheated battery pack met the power demand of external load. The controlled temperature in the temperature holding was increased stepwise, which was most energy-efficient compared with fixing the controlled temperature at 0°C , 5°C , and 10°C .
- (4) Results showed that the battery pack could be rapidly heated from -41.4°C to -14.0°C in 371 s by consuming 6.1% of nominal capacity without being over-discharged. Moreover, the strategy allowed the battery pack to output 64.81% and 66.59% of nominal energy for loading the modified DST and the modified FUDS respectively after preheating.

The limitation of the strategy is that the initialization parameters used in the process switch are specified according to the results of repetitive tests. Future research will focus on optimizing the initialization parameters adaptively based on to both the historical data and the real-time data of dynamic loading.

Credit author statement

Deyang Huang: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation, Resources. Ziqiang Chen: Investigation, Supervision, Writing – review & editing, Project administration, Funding acquisition. Shiyao Zhou: Investigation, Review, Validation.

Declaration of competing interest

The authors declare that they have no known competing

financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

TABLE A1

Summary of EKF algorithm for SoC estimation

0. Definition:
Establish state vector and matrices based on Eq [1]. ~ [6]:
$\mathbf{x}_i(k) = \begin{bmatrix} S_{oc,i}^{cell}(k) & V_{p,i}^{cell}(k) \end{bmatrix}$, $u_i(k) = I_{pack}^{pack}(k)$, $y_i(k) = V_{t,i}^{cell}(k)$
$\mathbf{A}_i(k) = \begin{bmatrix} 1 & 0 \\ 0 & e^{-\Delta t_e/\tau_i(k)} \end{bmatrix}$, $\mathbf{B}_i(k) = \begin{bmatrix} \eta_c \Delta t_e / 3600 C_a \\ R_{p,i}^{cell}(k)(1 - e^{-\Delta t_e/\tau_i(k)}) \end{bmatrix}$, $\mathbf{C}_i(k) = \begin{bmatrix} \frac{df(S_{oc,i}^{cell})}{dS_{oc,i}^{cell}} \big _{S_{oc,i}^{cell} = S_{oc,i}^{cell-}(k)} & 1 \end{bmatrix}$
1. Initialization at $k = 0$
$\hat{\mathbf{x}}_i^+(0) = E[\mathbf{x}_i(0)]$, $\mathbf{P}_{ekf,i}^+(0) = E[(\mathbf{x}_i(0) - \hat{\mathbf{x}}_i^+(0))(\mathbf{x}_i(0) - \hat{\mathbf{x}}_i^+(0))^T]$, $\mathbf{Q}$, $\mathbf{R}$
where $\mathbf{Q}$ and $\mathbf{R}$ are the covariance matrices of the independent, zero-mean, Gaussian noise.
2. Time updating from $(k-1)^+$ to k^- for prior estimation ($k = 1, 2, \dots$):
$\hat{\mathbf{x}}_i^-(k) = \mathbf{A}_i(k-1)\hat{\mathbf{x}}_i^+(k-1) + \mathbf{B}_i(k-1)u_i(k-1)$
$\mathbf{P}_{ekf,i}^-(k) = \mathbf{A}_i(k)\mathbf{P}_{ekf,i}^+(k-1)\mathbf{A}_i(k)^T + \mathbf{Q}$
$e_{ekf,i}(k) = y_i(k) - (f(\hat{S}_{oc,i}^{cell-}(k)) + R_{o,i}^{cell}(k)u_i(k) + \hat{V}_{p,i}^{cell-}(k))$
3. Measurement updating from k^- to k^+ for posterior estimation:
Calculate Kalman gain: $\mathbf{K}_{ekf,i}(k) = \mathbf{P}_{ekf,i}^-(k)\mathbf{C}_i(k)^T(\mathbf{C}_i(k)\mathbf{P}_{ekf,i}^-(k)\mathbf{C}_i(k)^T + \mathbf{R})^{-1}$
Update state vector: $\hat{\mathbf{x}}_i^+(k) = \hat{\mathbf{x}}_i^-(k) + \mathbf{K}_{ekf,i}(k)(y_i(k) - e_{ekf,i}(k))$
Update covariance matrix of estimation: $\mathbf{P}_{ekf,i}^+(k) = (\mathbf{I} - \mathbf{K}_{ekf,i}(k)\mathbf{C}_i(k))\mathbf{P}_{ekf,i}^-(k)$
4. Return to step 2 for next time updating

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